System Documentation

Electronics

BlueLine
with Multifunction Display MFD
Series 2000 (CR) FPP
ADEC
Application: Marine
Functional Description
Operating Instructions
Workshop Manual
Installation and
Commissioning Instructions

E532776/03E

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1 Safety

1.1 Important requirements for all products

Nameplate

The product is identified by a nameplate, model designation or serial number which must match the information on the title page of this manual.

Nameplate, model designation or serial number can be found on the product.

General information

This product may pose a risk of injury or damage in the following cases:
- Improper use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications
- Noncompliance with the safety instructions and warning notices

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:
- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/MTU contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (including engine management/parameters)
- In compliance with all safety regulations and in accordance with all warning notices in this manual
- With maintenance work performed in accordance with the (→ Maintenance Schedule) throughout the useful life of the product
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair
- By contracting only workshops authorized by the manufacturer to carry out repair and overhaul

Any other use shall be considered non-intended. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

Changes or modifications

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

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1.2 Personnel and organizational requirements

Organizational measures of the operator

This manual must be issued to all personnel involved in operation, maintenance, repair or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair and transport personnel at all times.

Use this manual as a basis for instructing personnel on product operation and repair, whereby the safety-relevant instructions, in particular, must be read and understood.

This is particularly important in the case of personnel who only occasionally perform work on or around the product. This personnel must be instructed repeatedly.

Personnel requirements

All work on the product shall be carried out by trained and qualified personnel only.

- Training at the Training Center of the manufacturer
- Qualified personnel specialized in mechanical and plant engineering

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair and transport.

Working clothes and personal protective equipment

Wear proper protective clothing for all work.

When working, always wear the necessary personal protective equipment (e.g. ear protectors, protective gloves, goggles, breathing protection). Observe the information on personal protective equipment in the respective activity description.

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1.3 Safety requirements for startup and operation

Safety requirements for startup

The product must be installed and accepted in accordance with manufacturers' specifications prior to initial startup.

All the requisite regulatory approvals must have been granted and all startup requirements fulfilled prior to initial startup.

Whenever the product is started, always ensure that:
- All maintenance and repair work has been completed.
- All loose parts have been removed from rotating machine components.
- Wearers of cardiac pacemakers or other active medical implants are well clear.
- The operating room is adequately ventilated.
- Exhaust pipework is leak-tight and routed to atmosphere.
- Battery terminals, generator terminals and cables are guarded to preclude accidental contact.
- All personnel is clear of the danger zone represented by moving parts.

Immediately after putting the product into operation, make sure that all control and display instruments as well as the monitoring, signaling and alarm systems are working properly.

Safety requirements for operation

The operator must be familiar with the controls and displays.

The operator must be aware of the consequences of any operations he/she performs.

During operation, the display instruments and monitoring units must be constantly observed in regard of present operating status, limit value violation and warning or alarm messages.

Malfunctions and emergency stop

Emergency procedures, in particular emergency stop, must be practiced on a regular basis.

Take the following steps if a system malfunction is detected or signaled by the system:
- Inform the duty supervisor(s).
- Evaluate the message.
- Respond to the emergency appropriately, e.g. execute an emergency stop.

Operation

Do not stay in the operating room when the product is running unless absolutely necessary and then as briefly as possible.

Keep a safe distance away from the product whenever possible. Do not touch the product unless expressly instructed to do so.

Do not inhale exhaust gases emitted by the product.

Ensure that the following requirements have been fulfilled before starting the product:
- Ear protectors are worn.
- Mop up any leaked or spilled fluids and lubricants immediately or soak up with a suitable binding agent.

Operation of electrical equipment

Some components are live (high voltage) when electrical equipment is in operation.

Observe the safety instructions for these appliances.

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1.4 Safety requirements for maintenance and repair work

Safety requirements before commencing maintenance and repair work

Have maintenance or repair work carried out by qualified and authorized personnel only.

Allow the product to cool down to less than 50°C (risk of explosion for oil vapors, fluids and lubricants, risk of burning).

Relieve pressure in fluid and lubricant systems and compressed-air lines which are to be opened. Use suitable collection vessels with a sufficient filling volume.

Ensure that the operating room is adequately ventilated when changing the oil or working on the fuel system.

Do not perform maintenance or repair work when the product is running unless:
- expressly instructed to do so.
- the product is running in the low load range and only for as long as necessary to complete the task.

Lock-out the product to preclude undesired starting, e.g. start interlock.

Tag-out the product with a “Do Not Start” sign in the operation room or on the control facility.

Disconnect the battery. Lock the contactor.

Close the main valve on the compressed-air system and vent the compressed-air line when pneumatic starters are fitted.

Disconnect the control facility from the product.

For starters with pinions made of copper-beryllium alloy:
- Wear a respirator mask (filter class P3). Do not blow out the interior of the flywheel housing or the starter with compressed air. Clean the interior of the flywheel housing with a class H dust extractor.
- Observe the safety data sheet.

Safety requirements during maintenance and repair work

Take special care when removing vent plugs or plug screws from the product. Hold a cloth over the screw or plug to prevent discharge of highly pressurized liquids.

Take care when draining hot fluids and lubricants (risk of burning).

Use suitable and calibrated tools only. Observe the specified tightening torques during assembly or disassembly.

Carry out work only on assembles and/or installations which are properly secured.

Never climb up on the lines.

Keep fuel injection lines and connections clean.

Always seal connections with caps or covers if a line is removed or opened.

Fit new seals when re-installing lines.

Avoid damaging lines, particularly the fuel lines.

Ensure that all retainers and dampers are installed correctly.

Ensure that all fuel injection and pressurized oil lines are installed with enough clearance to prevent contact with other components. Never route fuel or oil lines in the vicinity of hot components.

Do not touch elastomeric seals (e.g. Viton sealing rings) with your bare hands if they have a carbonized or resinous appearance.

Observe the cooling time for components which have been heated for installation or removal (risk of burning!).

Always use suitable ladders and work platforms when working above head-height. Ensure that components or assemblies are placed on stable surfaces.

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Pay particular attention to cleanliness at all times.

Safety requirements after completing maintenance and repair work

Ensure that all personnel is clear of danger zones before cranking.

Check that all access ports/apertures which have been opened to facilitate working are closed again.

Check that all safety equipment has been installed and that all tools and loose parts have been removed (especially the barring gear).

Ensure that no unattached parts have been left in/on the product (e.g. including rags and cable straps).

Welding work

Do not perform welding on the product or its attachments. Cover the product when performing welding work in the vicinity.

Before commencing welding work:
- Switch off the master power supply switch.
- Disconnect the battery.
- Disconnect electronic and genset grounds.

Do not perform maintenance or repair work on the product when welding is in progress in its vicinity. Risk of explosion or fire due to oil vapors and highly flammable process materials.

Do not use the product as a grounding terminal.

Do not route the welding cable over or near the wiring harnesses of the product. The welding current may otherwise induce an interference voltage in the wiring harnesses which could conceivably damage the electrical system.

Remove components (e.g. exhaust pipe) from the product before performing necessary welding work .

Hydraulic installation and removal

Check function and satisfactory condition of the jigs and fixtures to be used. Use only the specified jigs and fixtures for hydraulic removal/installation.

Observe the max. permissible force-on pressure specified for the installation/removal jig.

Do not attempt to bend or exert force on HP lines.

Before starting work, pay attention to the following:
- Vent the installation/removal jig, the pumps and the pipework at the relevant designated points.
- For hydraulic installation, screw on the jig with the piston retracted.
- For hydraulic removal, screw on the jig with the piston extended.

For hydraulic installation/removal jigs with central expansion pressure supply, screw the spindle into the shaft end until correct sealing has been established.

During hydraulic installation and removal, ensure that nobody is standing in the immediate vicinity of the component to be installed/removed.

Working with batteries

Observe the safety requirements of the battery manufacturer when working with batteries.

Gases released from the battery are explosive. Avoid sparks and naked flames.

Do not allow electrolyte (battery acid) to come into contact with skin or clothing.

Wear goggles and protective gloves.

Do not place tools on the battery.

Check polarity before connecting the cable to the battery. Battery polarity reversal may lead to injury by the sudden discharge of acid or bursting of the battery unit.

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